Network Assessment: Basic Network with Firewall and Load Balancer

1. Objective

The objective of this assessment is to design, implement, and secure a basic company network in Cisco Packet Tracer. The network will connect two departments (Sales and HR) to their respective servers, provide load balancing simulation for a shared web service, and implement firewall rules to control access between departments while allowing web service access.

2. Network Topology & IP Addressing Plan

2.1 Logical Topology Diagram

The network consists of the following components connected as described:

- **Router (e.g., Cisco 2911):** Connected to the ASA Firewall's outside interface.
- **ASA 5505 Firewall:** Placed between the Router and the Switch.
    - Ethernet0/0 connects to the Router.
    - Ethernet0/1 connects to the Switch for the Sales VLAN.
    - Ethernet0/2 connects to the Switch for the HR VLAN.
- **Switch (e.g., Cisco 2960):** Connects to the ASA Firewall and all End Devices.
    - GigabitEthernet0/1 connects to ASA Ethernet0/1 (Sales VLAN).
    - GigabitEthernet0/2 connects to ASA Ethernet0/2 (HR VLAN).
    - FastEthernet0/1-3 connect Sales PCs, Sales Server, and Web Server 1.
    - FastEthernet0/4-6 connect HR PCs, HR Server, and Web Server 2.
- **End Devices:**
    - 2 x Sales PCs
    - 1 x Sales Server (dedicated)
    - 1 x Web Server 1 (shared web service)
    - 2 x HR PCs
    - 1 x HR Server (dedicated)
    - 1 x Web Server 2 (shared web service)

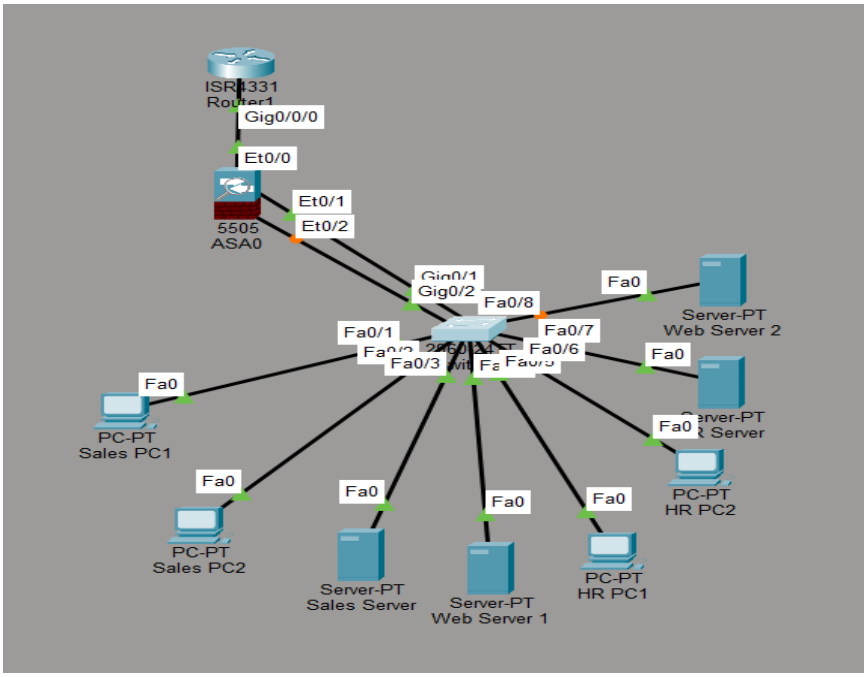


Fig 1: The network connections

2.2 IP Addressing Scheme

Device	Interface	IP Address	Subnet Mask	Default Gateway (if applicable)
Router	G0/0/0	10.0.0.1	255.255.255.0	N/A
ASA 5505	Vlan2 (outside)	10.0.0.2	255.255.255.0	N/A
	Vlan1 (sales)	192.168.1.1	255.255.255.0	N/A
	Vlan3 (hr)	192.168.2.1	255.255.255.0	N/A
Sales PC1	FastEthernet0	192.168.1.10	255.255.255.0	192.168.1.1
Sales PC2	FastEthernet0	192.168.1.11	255.255.255.0	192.168.1.1
Sales Server	FastEthernet0	192.168.1.50	255.255.255.0	192.168.1.1

Web Server 1	FastEthernet0	192.168.1.100	255.255.255.0	192.168.1.1
HR PC1	FastEthernet0	192.168.2.10	255.255.255.0	192.168.2.1
HR PC2	FastEthernet0	192.168.2.11	255.255.255.0	192.168.2.1
HR Server	FastEthernet0	192.168.2.50	255.255.255.0	192.168.2.1
Web Server 2	FastEthernet0	192.168.2.100	255.255.255.0	192.168.2.1

---

3. Device Configuration Steps

Follow these steps precisely, paying attention to the command mode (indicated by the prompt, e.g., Router(config)#).

3.1 Router Configuration

1. Open the Router's CLI.
2. Enter privileged EXEC mode, then global configuration mode.

Code:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.0
Router(config-if)#no shutdown


Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up


%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
```

Router(config-if)#exit

Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.2 # Route to Sales subnet via ASA

^

% Invalid input detected at '^' marker.

Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.2

Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2

Router(config)#end

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router#write memory

Building configuration...

[OK]

Router#

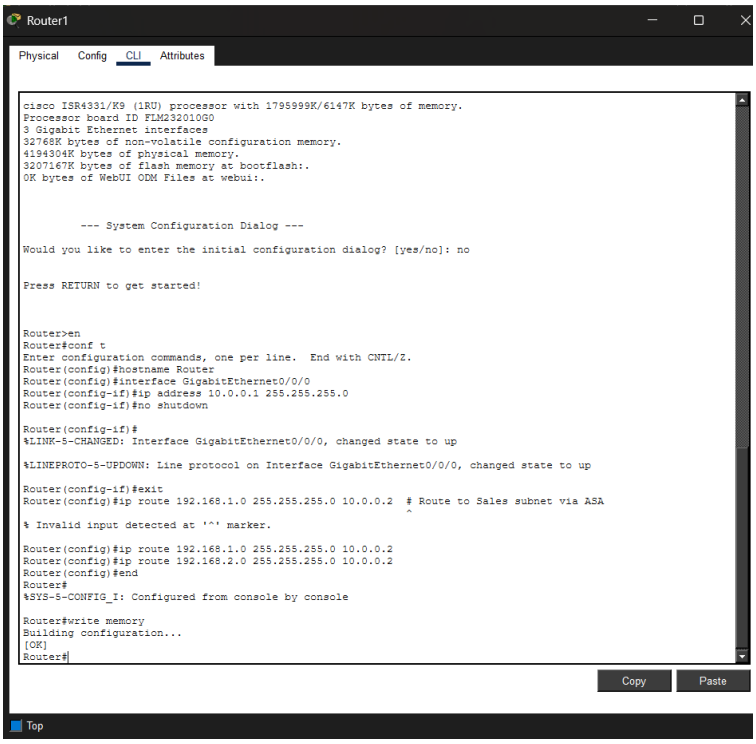


Fig 2: The CLI of the Router being configured.

3.2 Switch Configuration

1. Open the Switch's CLI.
2. Enter privileged EXEC mode, then global configuration mode.

Code:

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname Switch
Switch(config)#vlan 10
Switch(config-vlan)#name SALES_VLAN
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name HR_VLAN
Switch(config-vlan)#exit
Switch(config)#interface GigabitEthernet0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface GigabitEthernet0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface range FastEthernet0/1-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#no shutdown
Switch(config-if-range)#exit
```

Switch(config)#interface range FastEthernet0/5-8

Switch(config-if-range)#switchport mode access

Switch(config-if-range)#switchport access vlan 20

Switch(config-if-range)#no shutdown

Switch(config-if-range)#exit

Switch(config)#end

Switch#

%SYS-5-CONFIG_I: Configured from console by console

Switch#copy run start

Destination filename [startup-config]?

Building configuration...

[OK]

Switch#

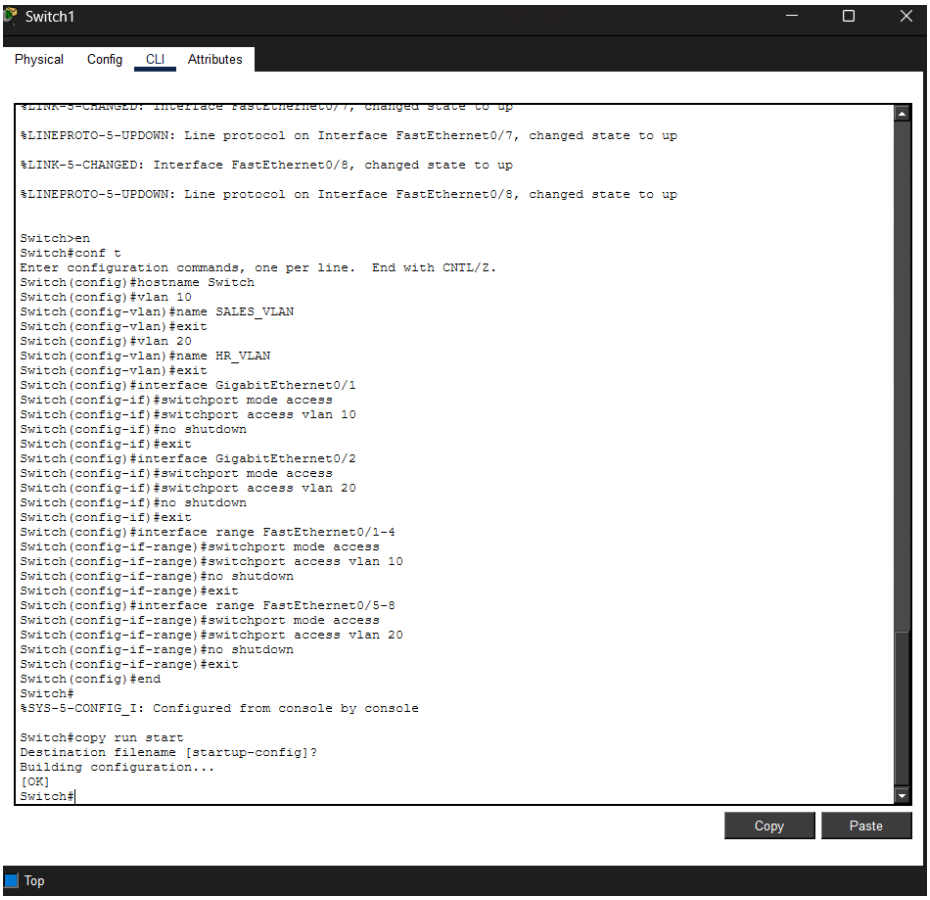


Fig 3: The CLI of the Switch being configured.

3.3 ASA 5505 Firewall Configuration

Important Note on ASA 5505 Base License: The ASA 5505 base license only allows 2 interfaces with nameif to be routable without requiring a no forward command. To configure a third nameif interface (our hr interface), we *must* use no forward interface Vlan1 (or Vlan2). This command means Vlan3 (hr) will *not* be able to route directly to Vlan1 (sales). This will impact HR's ability to reach Sales' web server.

1. Open the ASA's CLI.
2. Enter privileged EXEC mode, then global configuration mode.

Code:

```
ciscoasa>en

Password:

ciscoasa#conf t

ciscoasa(config)#hostname ASA

ASA(config)#interface Vlan2

ASA(config-if)#nameif outside

ASA(config-if)#security-level 0

ASA(config-if)#ip address 10.0.0.2 255.255.255.0

Waiting for the earlier webvpn instance to terminate...

Previous instance shut down.Starting a new one

ASA(config-if)#no shut

ASA(config-if)#exit

ASA(config)#interface Vlan1

ASA(config-if)#nameif sales

ASA(config-if)#security-level 100

ASA(config-if)#ip address 192.168.1.1 255.255.255.0

ASA(config-if)#no shut

ASA(config-if)#exit

ASA(config)#interface Vlan3

ASA(config-if)#no forward interface Vlan1

ASA(config-if)#nameif hr
```

```
INFO: Security level for "hr" set to 0 by default.
ASA(config-if)#security-level 50
ASA(config-if)#ip address 192.168.2.1 255.255.255.0
Waiting for the earlier webvpn instance to terminate...
Previous instance shut down.Starting a new one
ASA(config-if)#no shut
ASA(config-if)#exit
ASA(config)#interface Ethernet0/0
ASA(config-if)#switchport access vlan 2
ASA(config-if)#no shut
ASA(config-if)#exit
ASA(config)#interface Ethernet0/1
ASA(config-if)#switchport access vlan 1
ASA(config-if)#no shut
ASA(config-if)#exit
ASA(config)#interface Ethernet0/2
ASA(config-if)#switchport access vlan 3
ASA(config-if)#no shut
ASA(config-if)#exit
ASA(config)#class-map inspection_default
ASA(config-cmap)#match default-inspection-traffic
ASA(config-cmap)#exit
ASA(config)#policy-map global_policy
ASA(config-pmap)#class inspection_default
ASA(config-pmap-c)#inspect icmp
ASA(config-pmap-c)#inspect http
ASA(config-pmap-c)#exit
ASA(config)#exit
ASA#service-policy global_policy global
```

```
    ^
% Invalid input detected at '^' marker.

ASA#route outside 0.0.0.0 0.0.0.0 10.0.0.1 1
    ^
% Invalid input detected at '^' marker.

ASA#
ASA#conf t
ASA(config)#service-policy global_policy global
ASA(config)#route outside 0.0.0.0 0.0.0.0 10.0.0.1 1
ASA(config)#access-list HR_FILTER extended deny ip 192.168.2.0 255.255.255.0
192.168.1.50 255.255.255.255
ASA(config)#access-list HR_FILTER extended permit ip any any
ASA(config)#access-group HR_FILTER in interface hr
ASA(config)#access-list SALES_FILTER extended deny ip 192.168.1.0 255.255.255.0
192.168.2.50 255.255.255.255
ASA(config)#access-list SALES_FILTER extended permit ip any any
ASA(config)#access-group SALES_FILTER in interface sales
ASA(config)#write memory
Building configuration...
Cryptochecksum: 19547070 5d34790a 620a2f87 4df71c39

1383  bytes copied in 1.551 secs (891 bytes/sec)
[OK]
ASA(config)#end
ASA#
```

ASA0

Physical Config CLI Attributes

```
ASA(config)#interface Ethernet0/0
ASA(config-if)#switchport access vlan 2
ASA(config-if)#no shut
ASA(config-if)#exit
ASA(config)#interface Ethernet0/1
ASA(config-if)#switchport access vlan 1
ASA(config-if)#no shut
ASA(config-if)#exit
ASA(config)#interface Ethernet0/2
ASA(config-if)#switchport access vlan 3
ASA(config-if)#no shut
ASA(config-if)#exit
ASA(config)#class-map inspection_default
ASA(config-cmap)#match default-inspection-traffic
ASA(config-cmap)#exit
ASA(config)#policy-map global_policy
ASA(config-pmap)#class inspection_default
ASA(config-pmap-c)#inspect icmp
ASA(config-pmap-c)#inspect http
ASA(config-pmap-c)#exit
ASA(config)#exit
ASA#service-policy global_policy global
      ^
% Invalid input detected at '^' marker.

ASA#route outside 0.0.0.0 0.0.0.0 10.0.0.1 1
      ^
% Invalid input detected at '^' marker.

ASA#
ASA#conf t
ASA(config)#service-policy global_policy global
ASA(config)#route outside 0.0.0.0 0.0.0.0 10.0.0.1 1
ASA(config)#access-list HR_FILTER extended deny ip 192.168.2.0 255.255.255.0 192.168.1.50 255.255.255.255
ASA(config)#access-list HR_FILTER extended permit ip any any
ASA(config)#access-group HR_FILTER in interface hr
ASA(config)#access-list SALES_FILTER extended deny ip 192.168.1.0 255.255.255.0 192.168.2.50 255.255.255.255
ASA(config)#access-list SALES_FILTER extended permit ip any any
ASA(config)#access-group SALES_FILTER in interface sales
ASA(config)#write memory
Building configuration...
Cryptochecksum: 19547070 5d34790a 620a2f87 4df71c39

1383  bytes copied in 1.551 secs (891 bytes/sec)
[OK]
ASA(config)#end
ASA#
```

Copy Paste

Top

Fig 4: The CLI of the ASA Firewall being configured.

3.4 End Device (PC & Server) IP Configuration

For each PC and Server:

1. Click on the device in Packet Tracer.
2. Go to the **Desktop** tab.
3. Click on **IP Configuration**.
4. Enter the **IP Address**, **Subnet Mask** (255.255.255.0), and **Default Gateway** as per the IP Addressing Plan (Section 2.2).
5. Close the window (settings are saved automatically).

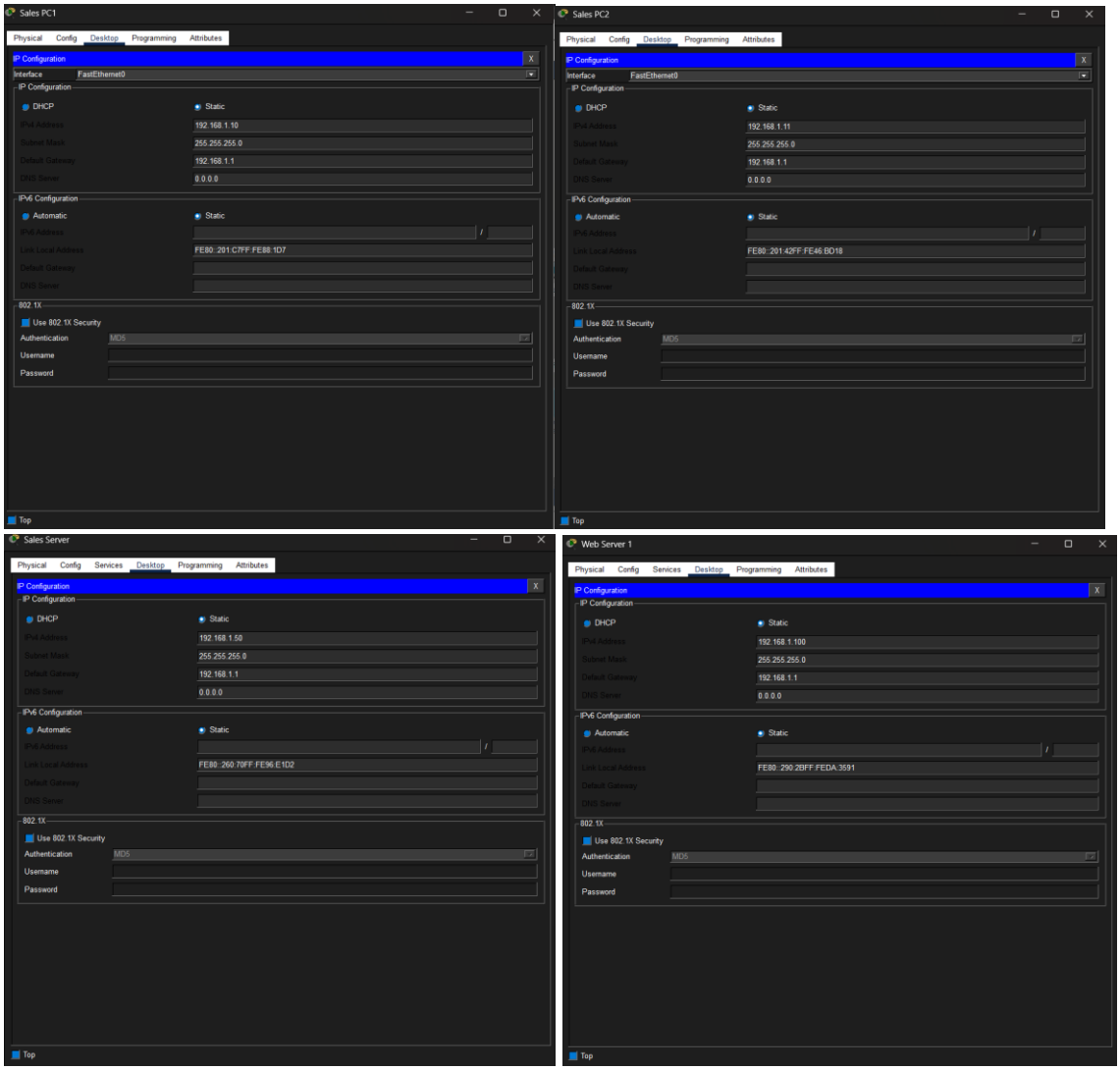


Fig 5: The assigning of IP to sales department.

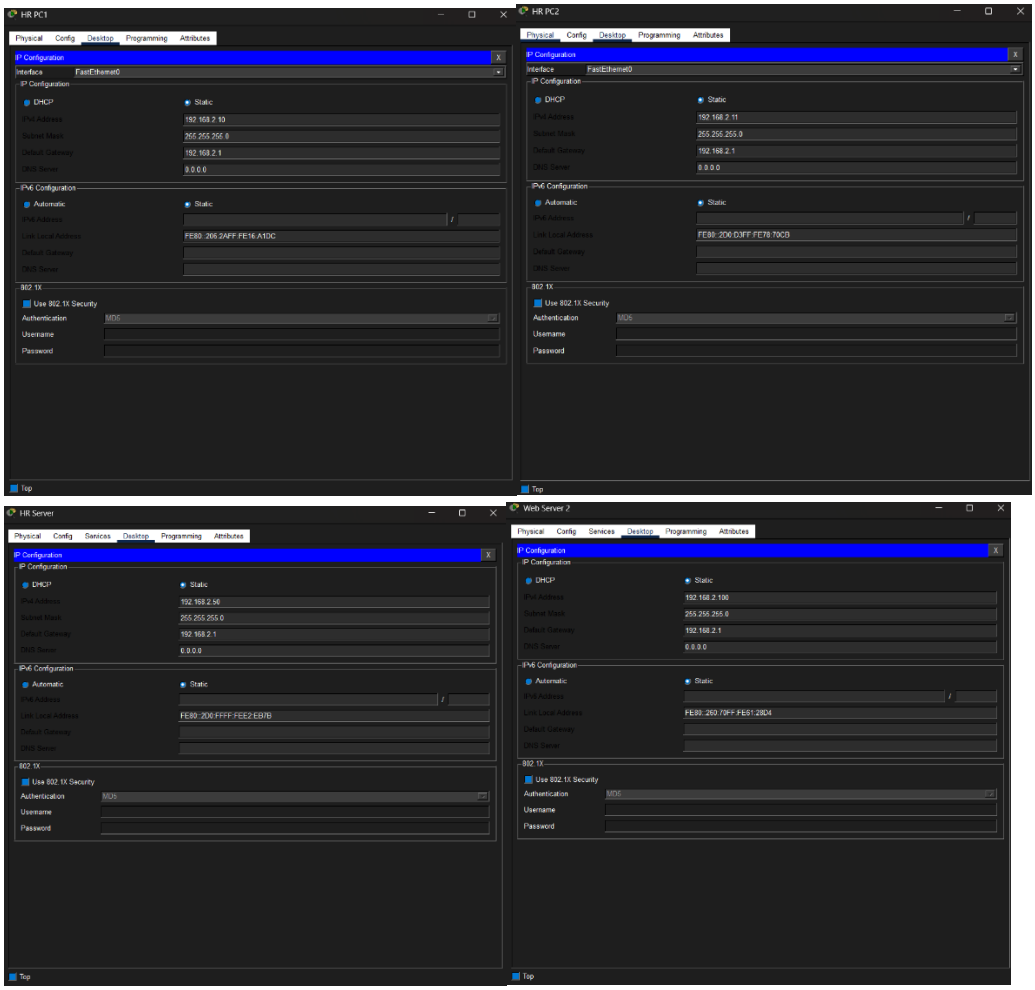


Fig 6: The assigning of IP to HR department.

3.5 Web Server Service & Custom Page Configuration

For both Web Server 1 and Web Server 2:

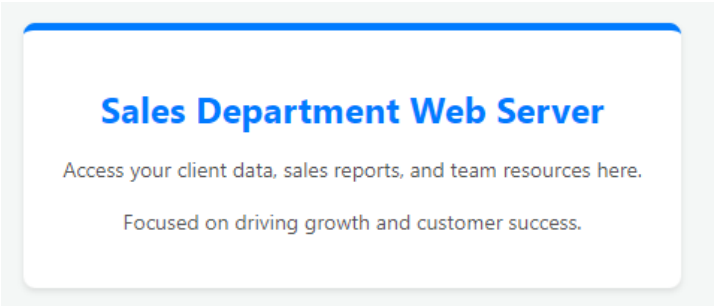
1. Click on the web server.
2. Go to the **Services** tab.
3. Click on **HTTP** in the left menu.
4. Ensure **HTTP** and **HTTPS** services are On.
5. Click on the index.html file, then click **Edit**.
6. Delete existing content and paste the following HTML code for the respective server.
7. Click **Save**.

For Web Server 1 (Sales Department - 192.168.1.100):

codeHtml

```
<!DOCTYPE html>
<html>
<head>
  <title>Sales Department Portal</title>
  <style>
    body {
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      background-color: #f4f7f6;
      margin: 0;
      padding: 20px;
      color: #333;
      text-align: center;
    }
    .container {
      background-color: #ffffff;
      border-radius: 8px;
      box-shadow: 0 2px 4px rgba(0,0,0,0.1);
```

```
        padding: 20px;
        max-width: 400px;
        margin: 20px auto;
        border-top: 5px solid #007bff; /* Sales accent color */
    }
    h1 {
        color: #007bff; /* Sales accent color */
        font-size: 24px;
        margin-bottom: 10px;
    }
    p {
        font-size: 14px;
        line-height: 1.5;
        color: #555;
    }
  </style>
</head>
<body>
  <div class="container">
    <h1>Sales Department Web Server</h1>
    <p>Access your client data, sales reports, and team resources here.</p>
    <p>Focused on driving growth and customer success.</p>
  </div>
</body>
</html>
```



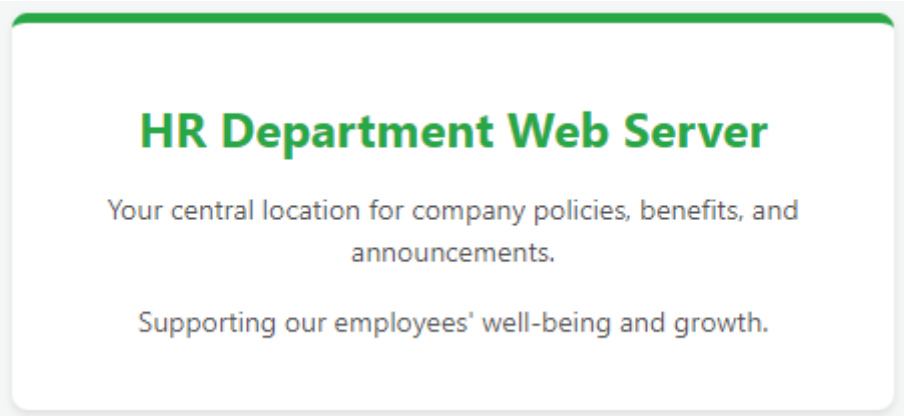
Preview1: Sales website

For Web Server 2 (HR Department - 192.168.2.100):

codeHtml

```
<!DOCTYPE html>
<html>
<head>
  <title>HR Employee Hub</title>
  <style>
    body {
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      background-color: #f4f7f6;
      margin: 0;
      padding: 20px;
      color: #333;
      text-align: center;
    }
    .container {
      background-color: #ffffff;
      border-radius: 8px;
      box-shadow: 0 2px 4px rgba(0,0,0,0.1);
      padding: 20px;
      max-width: 400px;
      margin: 20px auto;
      border-top: 5px solid #28a745; /* HR accent color */
```

```
        }
        h1 {
            color: #28a745; /* HR accent color */
            font-size: 24px;
            margin-bottom: 10px;
        }
        p {
            font-size: 14px;
            line-height: 1.5;
            color: #555;
        }
    </style>
</head>
<body>
    <div class="container">
        <h1>HR Department Web Server</h1>
        <p>Your central location for company policies, benefits, and announcements.</p>
        <p>Supporting our employees' well-being and growth.</p>
    </div>
</body>
</html>
```



Preview2: HR website

4. Connectivity Testing & Verification

Perform the following tests from the PCs to verify connectivity and firewall rules. Take screenshots of these results for your submission.

4.1 Testing from Sales PCs (e.g., Sales PC1 - 192.168.1.10)

Open Desktop -> Command Prompt on Sales PC1.

1. **Ping Sales Server (192.168.1.50)**
    - **Command:** ping 192.168.1.50
    - **Expected Output:** Successful Replies (Reply from 192.168.1.50...).
    - **Reason:** Within the same subnet, allowed by default and by ASA security level rules (same security level traffic is allowed).
2. **Ping Web Server 1 (192.168.1.100)**
    - **Command:** ping 192.168.1.100
    - **Expected Output:** Successful Replies (Reply from 192.168.1.100...).
    - **Reason:** Within the same subnet, allowed.
3. **Ping HR Server (192.168.2.50)**
    - **Command:** ping 192.168.2.50
    - **Expected Output:** Request timed out or Destination Host Unreachable.
    - **Reason:** Blocked by the SALES_FILTER ACL applied inbound on the sales interface (access-list SALES_FILTER extended deny ip 192.168.1.0 255.255.255.0 host 192.168.2.50).
4. **Ping Web Server 2 (192.168.2.100)**
    - **Command:** ping 192.168.2.100
    - **Expected Output:** Successful Replies (Reply from 192.168.2.100...).
    - **Reason:** Sales (security-level 100) can initiate connections to HR (security-level 50) by default. The SALES_FILTER ACL only specifically denies access to the HR Server (192.168.2.50), not the HR Web Server.
5. **Access Web Server 1 via Web Browser**
    - **Action:** Go to Desktop -> Web Browser, type 192.168.1.100 in the URL bar, and click Go.
    - **Expected Output:** The custom "Sales Department Web Server" page (blue theme).
    - **Reason:** Direct access within its subnet.

6. **Access Web Server 2 via Web Browser**
    - **Action:** Go to Desktop -> Web Browser, type 192.168.2.100 in the URL bar, and click Go.
    - **Expected Output:** The custom "HR Department Web Server" page (green theme).
    - **Reason:** Sales to HR traffic is allowed (higher to lower security level), and not blocked by ACL for this specific IP.

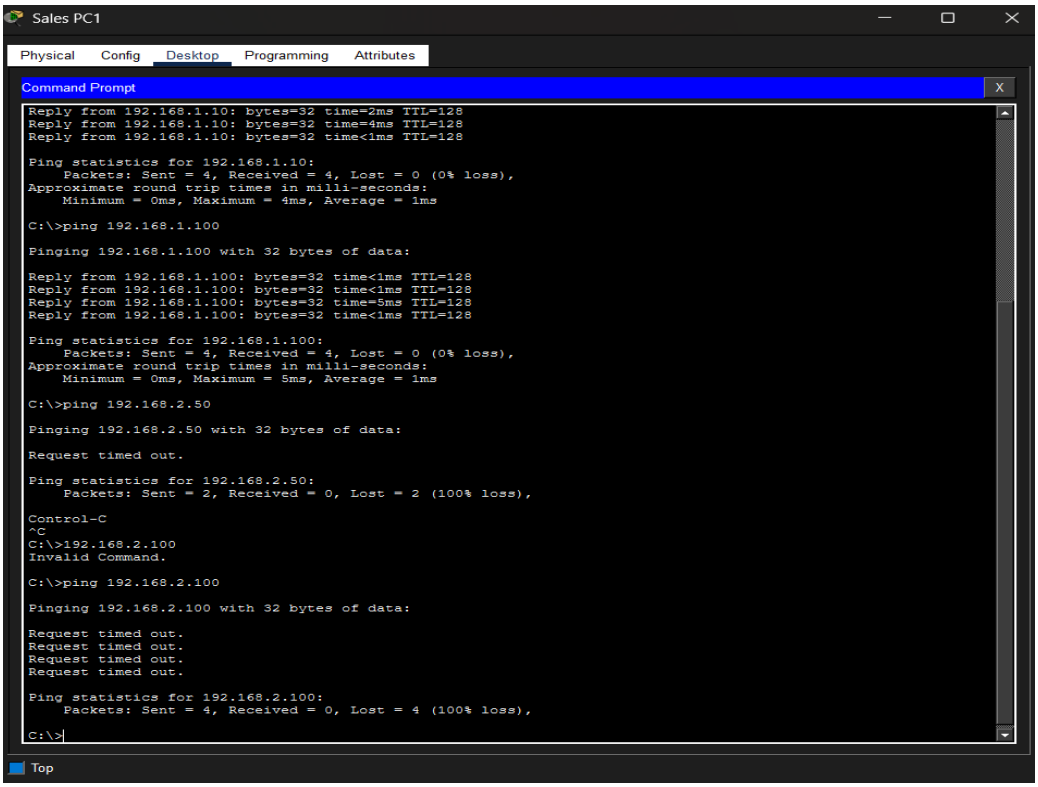


Fig 7: The Output at Sales PC-1.

4.2 Testing from HR PCs (e.g., HR PC1 - 192.168.2.10)

Open Desktop -> Command Prompt on HR PC1.

1. **Ping HR Server (192.168.2.50)**
    - **Command:** ping 192.168.2.50
    - **Expected Output:** Successful Replies (Reply from 192.168.2.50...).
    - **Reason:** Within the same subnet, allowed.
2. **Ping Web Server 2 (192.168.2.100)**
    - **Command:** ping 192.168.2.100
    - **Expected Output:** Successful Replies (Reply from 192.168.2.100...).

- **Reason:** Within the same subnet, allowed.

3. **Ping Sales Server (192.168.1.50)**
   - **Command:** ping 192.168.1.50
   - **Expected Output:** Request timed out or Destination Host Unreachable.
   - **Reason:** Blocked primarily by the ASA 5505's no forward interface Vlan1 command on Vlan3 (hr), which prevents direct routing from HR to Sales. Additionally, the HR_FILTER ACL explicitly denies this.
4. **Ping Web Server 1 (192.168.1.100)**
   - **Command:** ping 192.168.1.100
   - **Expected Output:** Request timed out or Destination Host Unreachable.
   - **Reason:** Blocked by the ASA 5505's no forward interface Vlan1 command on Vlan3 (hr), preventing HR from reaching any resource in the Sales subnet (192.168.1.x).
5. **Access Web Server 2 via Web Browser**
   - **Action:** Go to Desktop -> Web Browser, type 192.168.2.100 in the URL bar, and click Go.
   - **Expected Output:** The custom "HR Department Web Server" page (green theme).
   - **Reason:** Direct access within its subnet.
6. **Access Web Server 1 via Web Browser**
   - **Action:** Go to Desktop -> Web Browser, type 192.168.1.100 in the URL bar, and click Go.
   - **Expected Output:** Web browser will likely display "Request Timed Out" or similar error.
   - **Reason:** Blocked by the ASA 5505's no forward interface Vlan1 command on Vlan3.

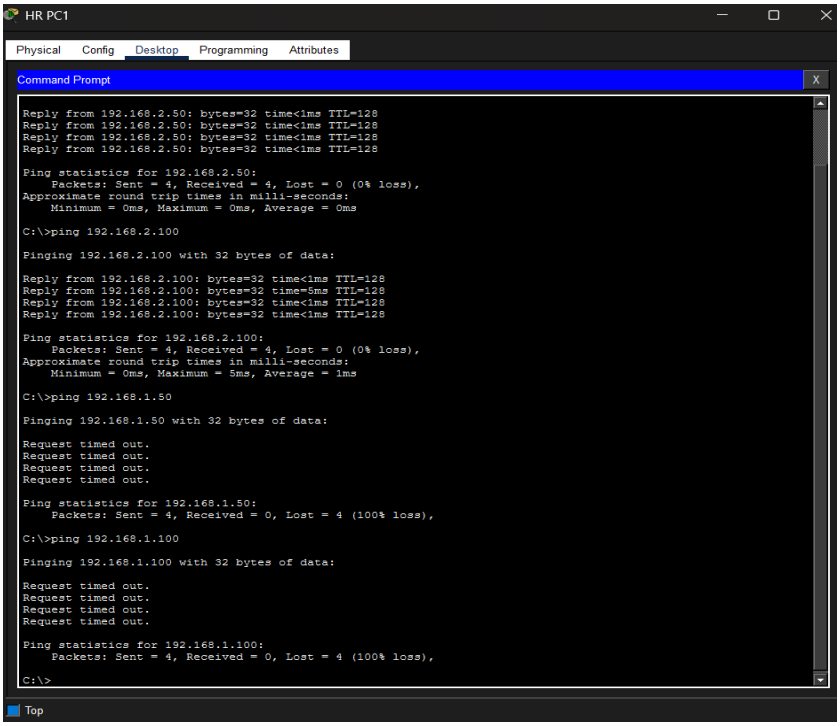


Fig 7: The Output at HR PC-1.

5. Summary of Firewall Rules and Load Balancer Simulation

Firewall Rules Explanation

The ASA 5505 firewall was configured with three logical interfaces:

1. outside (VLAN 2, security-level 0): Connecting to the Router and the external network.
2. sales (VLAN 1, security-level 100): Serving the Sales department (192.168.1.x) and Web Server 1.
3. hr (VLAN 3, security-level 50): Serving the HR department (192.168.2.x) and Web Server 2.

Key Security Mechanisms:

- **Security Levels:** By default, traffic can flow from a higher security level to a lower security level (e.g., sales to hr, sales to outside, hr to outside) without explicit ACLs, but not in the reverse direction unless explicitly permitted.
- **ASA 5505 no forward Limitation:** The command no forward interface Vlan1 was applied to the hr interface (VLAN 3). This is a license-imposed restriction that prevents *any* direct routing between VLAN 3 (HR) and VLAN 1 (Sales). This fundamentally blocks HR PCs from reaching Sales servers or Web Server 1, regardless of ACLs.

- **Access Control Lists (ACLs):**
  - HR_FILTER (applied inbound on hr interface): Explicitly denies traffic from the HR subnet (192.168.2.0/24) to the Sales Server (192.168.1.50). A permit ip any any statement allows all other HR-originated traffic (e.g., to the internet or Web Server 2).
  - SALES_FILTER (applied inbound on sales interface): Explicitly denies traffic from the Sales subnet (192.168.1.0/24) to the HR Server (192.168.2.50).
    A permit ip any any statement allows all other Sales-originated traffic (e.g., to the internet, Web Server 1, or Web Server 2).
- **Global Inspection Policy:** The global_policy with inspect icmp and inspect http ensures that these common protocols are correctly handled and statefully inspected by the firewall, allowing responses to return through the firewall.

Load Balancer Simulation

The "load balancer simulation" was achieved by deploying two identical web servers (Web Server 1 and Web Server 2) in separate subnets, each accessible by at least one department.

- Web Server 1 (192.168.1.100) is primarily accessible by the Sales department.
- Web Server 2 (192.168.2.100) is primarily accessible by the HR department.

Outcome:

- Sales PCs can access both Web Server 1 and Web Server 2.
- HR PCs can only access Web Server 2.
  This demonstrates a distributed web service, where users can connect to a functional web server. The ASA 5505's no forward limitation prevented full cross-department access to *both* web servers, which would typically be a goal for true load balancing. For a more complete load balancing scenario, a dedicated load balancer device or a different firewall model without such license restrictions would be employed.